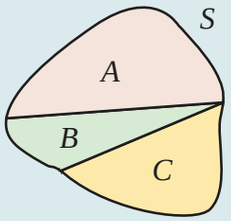
Solution

Since the experiment has exactly the three possible mutually exclusive outcomes A, B and C, they must be exhaustive events.

$$\Rightarrow S = A \cup B \cup C$$



Therefore, by axioms of probability

$P(A) \geq 0, \quad P(B) \geq 0, \quad P(C) \geq 0$ and

$P(A \cup B \cup C) = P(A) + P(B) + P(C) = P(S) = 1$

(i) Given that $P(A) = \frac{4}{7} \geq 0$, $P(B) = \frac{1}{7} \geq 0$, and $P(C) = \frac{2}{7} \geq 0$

Also $P(S) = P(A) + P(B) + P(C) = \frac{4}{7} + \frac{1}{7} + \frac{2}{7} = 1$

Therefore the assignment of probability is permissible.

(ii) Given that $P(A) = \frac{2}{5} \geq 0$, $P(B) = \frac{1}{5} \geq 0$, and $P(C) = \frac{3}{5} \geq 0$

But $P(S) = P(A) + P(B) + P(C) = \frac{2}{5} + \frac{1}{5} + \frac{3}{5} = \frac{6}{5} > 1$

Therefore the assignment is not permissible.

(iii) Since $P(C) = -0.2$ is negative, the assignment is not permissible.

(iv) The assignment is permissible because

$P(A) = \frac{1}{\sqrt{3}} \geq 0$, $P(B) = 1 - \frac{1}{\sqrt{3}} \geq 0$, and $P(C) = 0 \geq 0$

$P(S) = P(A) + P(B) + P(C) = \frac{1}{\sqrt{3}} + 1 - \frac{1}{\sqrt{3}} + 0 = 1.$

(v) Even though $P(A) = 0.421 \geq 0$, $P(B) = 0.527 \geq 0$, and $P(C) = 0.042 \geq 0$, the sum of the probability

$P(S) = P(A) + P(B) + P(C) = 0.421 + 0.527 + 0.042 = 0.990 < 1.$

Therefore, the assignment is not permissible.

Example 12.2

An integer is chosen at random from the first ten positive integers. Find the probability that it is (i) an even number (ii) multiple of three.

Solution

The sample space is

$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $n(S) = 10$

Let A be the event of choosing an even number and

B be the event of choosing an integer multiple of three.

$A = \{2, 4, 6, 8, 10\}$, $n(A) = 5$,

$B = \{3, 6, 9\}$, $n(B) = 3$

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